

Program : Diploma in Textile Technology	
Course Code : 5063A	Course Title: Non-wovens & Technical Textiles
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To familiarize the students with the manufacturing of Non-woven fabrics and the applications areas of technical textiles

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge of Textile Fibre		Textile Fibres	2
Basic knowledge in Yarn Manufacture		Yarn Manufacture I & II	3 & 4
Basic knowledge in Fabric Manufacture		Fabric Manufacture I & II	3 & 4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	To impart knowledge on Non-woven process and to implement the web forming methods.	14	Applying
CO2	Execute the different bonding and finishing techniques of non-wovens.	14	Applying
CO3	Summarise the concept and various sectors of technical textiles- Applications of Geo-textiles and Agro-textiles.	15	Applying
CO4	Application of technical textile in Medical and Industrial sectors	15	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					
CO3			3				
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To impart knowledge on Non-woven process and to implement the web forming methods.		
M1.01	Summarize the production process of Non-wovens and fibres used for non-woven production	7	Understanding
M1.02	To implement various web forming methods- Dry-laid, wet-laid and polymer-laid	7	Applying
Contents: Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods			
CO2	Execute the different bonding and finishing techniques of non-wovens		
M2.01	To know various Nonwoven bonding techniques	5	Understanding
M2.02	To execute the process of Mechanical, Chemical and Thermal bonding nonwovens and to test and compare parameters of each type.	5	Applying
M2.03	Discuss the Finishing and limitations of nonwovens	4	Understanding
	Series Test- I	1	

Contents:

Bonding methods: mechanical, thermal, chemical/ adhesive.

Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications.

Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens.

Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding.

Testing the fabric quality of each bonding method.

Finishing of bonded fabrics. Fusing - methods of fusing.

Discuss the limitations of non-woven fabrics.

CO3	Summarise the concept and various sectors of technical textiles- Applications of Geo-textiles and Agro-textiles.		
M3.01	Illustrate the scope and major applications of technical textiles	5	Understanding
M3.02	Application and analysis of Geotextiles	5	Applying
M3.03	Use of agro-textiles	5	Applying

Contents:

Technical Textiles: Definition - scope - State various applications of Technical Textiles.

Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications

Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications

CO4	Application of technical textile in Medical and Industrial sectors		
M4.01	Medical application of Technical Textiles	7	Applying
M4.02	Application of technical textile in industry sector	8	Applying
	Series Test - II	1	

Contents:

Medical Textiles: Classification and fibres used - requirements.

Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads.

Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics.

Bullet proof fabric construction and its strength analysis, Fire proof fabrics.

Text /Reference:

T/R	Book Title/Author
T1	A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile Institute
R2	Prof. N. N. Banerjee Nonwovens manufacture, 1979
R3	N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall
R4	T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, Structure, Properties and Applications- Textile institute(Wood head publishing India)
R5	R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, Vol.2, 1970 Textile institute
R6	V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow
R7	K P Chellaman i - Yarns& Technical Textiles SITRA Publication
R8	Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles NCUTE, IIT, New Delhi Delhi

Online resources:

Sl.No	Website Link
1	http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers-according/ (accessed on 20 October 2013)
2	https://nptel.ac.in/courses/116/102/116102026/
3	https://nptel.ac.in/courses/116/102/116102016/
4	https://swayam.gov.in/
5	http://eacharya.inflibnet.ac.in/vidya-mitra/